

Khaled Serag

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Education

Purdue University

Ph.D. in Computer Science

West Lafayette, Indiana, USA

August 2017 - August 2023

- **Thesis:** Proactive Vulnerability Identification and Defense Construction – The Case for CAN
- **Advisor:** Dongyan Xu
- **Co-advisor:** Z. Berkay Celik

State University of New York at Binghamton

M.S. in Electrical and Computer Engineering

Binghamton, New York, USA

December 2015

- **Specialization:** Information Assurance
- **GPA:** 3.91

Ain Shams University

B.S. in Electrical Engineering

Cairo, Egypt

September 2012

- **General Grade:** G.
- **Major Grade:** V.G.

Publications and Patents

ACADEMIC PAPERS

SoK: Kicking CAN Down the Road: Systematizing CAN Security Knowledge. Khaled Serag, Zhaozhou Tang, Sungwoo Kim, Vireshwar Kumar, Saman Zonouz, Raheem Beyah, Dongyan Xu, Z. Berkay Celik. *Network and Distributed System Security (NDSS) Symposium*, 2027.

A Formal Security Analysis of CAN XL (Distinguished Paper Award Runner-Up). Zhaozhou Tang, Khaled Serag, Z. Berkay Celik, Vijay Ganesh, Saman Zonouz, Raheem Beyah. *35th USENIX Security Symposium (USENIX Security 26)*, 2026.

WIP: Intrusion Detection and Localization for CAN by Extracting Propagation Delay Features from Message Intervals. Zhaozhou Tang, Khaled Serag, Saman Zonouz, Z. Berkay Celik, Dongyan Xu, Raheem Beyah. *3rd USENIX Symposium on Vehicle Security and Privacy*, 2025.

ERACAN: Defending Against an Emerging CAN Threat Model (Distinguished Paper Award). Zhaozhou Tang, Khaled Serag, Saman Zonouz, Z. Berkay Celik, Dongyan Xu, Raheem Beyah. *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2024.

ZBCAN: A Zero-Byte CAN Defense System. Khaled Serag, Rohit Bhatia, Akram Faqih, Muslum Ozgur Ozmen, Vireshwar Kumar, Z. Berkay Celik, Dongyan Xu. *32nd USENIX Security Symposium (USENIX Security 23)*, 2023.

Attacks on CAN Error Handling Mechanism. Khaled Serag, Vireshwar Kumar, Z. Berkay Celik, Rohit Bhatia, Mathias Payer, Dongyan Xu. *Fourth International Workshop on Automotive and Autonomous Vehicle Security (AutoSec)*, 2022.

Exposing New Vulnerabilities of Error Handling Mechanism in CAN. Khaled Serag, Rohit Bhatia, Vireshwar Kumar, Z. Berkay Celik, Dongyan Xu. *30th USENIX Security Symposium (USENIX Security 21)*, 2021.

Evading Voltage-Based Intrusion Detection on Automotive CAN. Rohit Bhatia, Vireshwar Kumar, Khaled Serag, Z. Berkay Celik, Mathias Payer, Dongyan Xu. *Network and Distributed System Security (NDSS) Symposium*, 2021.

PATENTS

Multiple Security Level Monitor for Monitoring a Plurality of MIL-STD-1553 Buses with Multiple Independent Levels of Security.

Josh D Eckhardt, Thomas E Donofrio, Khaled Serag. United States Patent No.: US10685125B2, 2020

Bus data monitor. Josh D Eckhardt, Thomas E Donofrio, Khaled Serag. United States Patent No.: US10691573B2, 2020

System and Method of Monitoring Data Traffic on a MIL-STD-1553 Data Bus. Josh D Eckhardt, Thomas E Donofrio, Khaled Serag. United States Patent No.: US10467174B2, 2019

Fellowships

Emil Stefanov Fellowship

For domestic graduate students who specialize in security and show originality and creative thinking in research

2022

Purdue University

Research Experience

Qatar Computing Research Institute

Scientist

Doha, Qatar

January 2024-Present

ICS Threat Intelligence (Paper Submitted)

May 2024-Present

- Design an ICS threat intelligence framework for system-wide incident investigation, real-time detection, and attack-path prediction
- Work with two engineers to design and deploy a multi-layer ICS testbed spanning enterprise, SCADA, control, and physical processes
- Work with two engineers to execute multi-stage ICS attacks and collect host, network, and physical-process telemetry
- Develop provenance-based methods to correlate evidence across ICS layers and reconstruct system-wide attack progression

10-BASE-T1S Vulnerability Identification

May 2024-Present

- Collaborate and offer advise to external researchers to identify vulnerabilities in 10-BASE-T1S Ethernet
- Collaborate and offer advise to external researchers to design a protocol fuzzing framework

Systematic Analysis of CAN Security (Two Papers Published)

August 2024-Present

- Create a systematic framework to organize CAN security literature into unified taxonomies and assessment models
- Use the framework to identify security gaps, trace their root causes, and assess their uniqueness to CAN
- Advise a student on formally modeling CAN XL and systematically identifying vulnerabilities in its MAC sub-layer
- Help design the formal analysis methodology and experimentally validate newly identified CAN XL vulnerabilities

Autonomous Vehicle Sensor Security

January 2025-Present

- Advise a student on formally analyzing vulnerabilities in autonomous vehicle sensor fusion algorithms
- Advise a student on modeling how sensor manipulation propagates through autonomous vehicle systems and affects vehicle behavior
- Apply assume-guarantee analysis to characterize system-level effects and safety violations caused by compromised sensor inputs

Purdue University

West Lafayette, Indiana, USA

Post-Doctoral Research Associate

August 2023 - January 2024

Intrusion Detection System for Emerging CAN Bus Attacker Models (Paper Published)

August 2023-January 2024

- Considering an emerging attacker model, analyze the new attacks they can launch and the improvements of established attacks
- Design a CAN intrusion detection system that considers the attacks enabled or improved under this new model

Anomaly Detection for Industrial Control Systems (ICS)

August 2023-January 2024

- Collaborate with team to devise a new approach to detecting ICS anomalies based on system invariants during normal operation
- Collaborate with team to put together an invariant extraction methodology that combines trace analysis and code analysis of PLCs

Purdue University

West Lafayette, Indiana, USA

Graduate Research Assistant

August 2017 - August 2023

Versatile and Performance-Friendly CAN Defense Construction (Paper Published)

January 2021-August 2023

- Design a CAN defense system that protects against the most common CAN attacks
- The system should have prevention and detection abilities
- The system should not use high-overhead operations such as encryption
- The system should not cause significant delays or significant busload increase and should not use message fields

CAN Error Handling Mechanism Vulnerability Identification (Multiple Papers Published)

August 2017-January 2021

- Identify vulnerabilities in CAN's error handling and fault confinement mechanism
- Showcase the different attack vectors that could take advantage of the discovered vulnerabilities
- Suggest ways to mitigate the discovered vulnerabilities
- Formalize and automate the vulnerability identification process
- Design a vulnerability scanning tool to test the protocol's error handling and fault confinement mechanism

Boeing

Huntsville, Alabama, USA

Software Engineer (Cybersecurity – Summer)

August 2017 - January 2022

Key Management for a Mesh-Networked Satellite System

May 2021 - June 2021

- Design a key management mechanism for a satellite network that uses mesh networking
- Provide forward and backward secrecy for nodes that join or leave the network
- Account for and differentiate between nodes that are joining and leaving the network and nodes that are being temporarily disconnected

Avionic CAN Bus Intrusion Detection System

May 2020-August 2020

- Make a list of the most common attack vectors for avionic CAN bus systems
- Elaborate on the pros on cons of different attack detection approaches, whether ML-based, Rule-based or hybrid
- Recommend features and rules to be watched for or enforced by any defense system

Vulnerability Assessment for a Mesh Network Using the Thread Protocol

May 2019 - August 2019

- Assess the security posture of the Thread protocol if implemented on cargo airplanes
- Assess the performance impact of Thread given a cargo airplane scenario
- Write a white paper listing the security and performance pros and cons if such implementation takes place

AFDX Switch Design and Analysis

May 2018 - August 2018

- Analyze current security threats to AFDX Systems
- Collaborate with team to design an AFDX switch with security measures to overcome the current security threats

Boeing

Huntsville, Alabama, USA

Software Engineer (Cybersecurity – Full Time)

February 2016 - August 2017

MIL-STD-1553 Guard/Monitor Design (Two Patents Published)

February 2016 - August 2017

- Collaborate with team to design a guard for MIL-STD-1553 systems using off-the-shelf components
- Investigate the impact of installing a guard between the protected payload and the bus on the system's latency
- Investigate the impact of installing a guard between the protected payload and the bus on the electrical characteristics of the system
- Study the different approaches to minimize the negative consequences of the guard installation on the system's latency

Multiple Independent Layers of Security for MIL-STD-1553 Systems (Patent Published)

September 2016 - August 2017

- Collaborate with team to secure multiple 1553 buses with different security levels running on a shared hardware
- Work with team to design an interface that maintains the separation between different security levels of each bus

Common Open Research Emulator (CORE) API Development

September 2016 - August 2017

- Collaborate with team to Develop CORE's API to incorporate more protocols/standards
- Investigate ways to develop a complete framework for wireless communications
- Develop CORE's software to facilitate the interaction between CORE and EMANE (Extendable Mobile Ad-hoc Network Emulator)

Threat Analysis for an Avionic System

January 2017 - July 2017

- Threat-analyze a system of TTTech Deterministic Ethernet, AFDX, and Windriver VxWorks
- Identify system assets, threat agents and system vulnerabilities
- Write a white paper containing a descriptive list of the possible attack vectors in addition to an attack tree

Network Monitoring Web Interface for Commercial Airplanes

April 2017 - July 2017

- Design a system to monitor real-time network performance metrics using Sflow
- Identify the relevant network metrics
- Collaborate with team to design a GUI web interface to show the network components and real-time metrics

Dynamic Network Resource Allocation for DARPA

June 2017 - July 2017

- Make a list of the specifications to adopt based on the Dynamic Link Exchange Protocol (DLEP)
- Update the Common Open Research Emulator's (CORE) source code to include Open vSwitch as a network element

State University of New York at Binghamton

Binghamton, New York, USA

Graduate Student Researcher

January 2014 - December 2015

Distributed Web Crawling System

September 2015 - December 2015

- Use Python to control Google Chrome browser, interface with pages, and gather data in real time
- Write a Crawling algorithm that allows for the specification of the crawling depth and the number of crawlers
- Create a database that collects the data gathered from running crawlers and keeps track of the visited URLs

Privacy Assurance on Facebook

January 2015 - May 2015

- Collaborate with my professor to find better ways to protect personal information on Facebook
- Collaborate with two students to use Steganography to embed secret pictures in cover pictures
- Investigate whether Partially Homomorphic Encryption (Additive, Multiplicative) could be beneficial if used to encrypt keys in the database

Dual Core Processor Design Using Verilog

April 2014 - May 2014

- Collaborate with two team-mates to design a simple dual core processor
- Write the code for the Caches, ALU's, and buffers, then synthesize the code using Synopsys

Other Professional Experience

Deloitte

New York City, New York, USA

Cyber Risk Intern

June 2015 - July 2015

- Collaborated with team to develop SIEM content for The State of Connecticut
- Created 8 Qradar reports based on 6 use cases

Security Meter

Giza, Egypt

Information Security Intern

September 2013 - December 2013

- Applied (SIEM) solutions for both Linux and windows computers of Banque Misr Using Qradar and Tenable
- Collaborated with 2 Engineers to apply Freeradius server and two-factor authentication (using Entrust) on the computers of Banque Misr

Talks and Presentations

Vulnerability Identification and Defense Construction in Cyber-Physical Systems

Presented at the Georgia Institute of Technology

January 2026

Georgia Tech, Atlanta

Vulnerability Identification and Defense Construction in Cyber-Physical Systems

Presented to the Center for Education and Research in Information Assurance and Security (CERIAS)

October 2023

Purdue University

ZBCAN: A Zero-Byte CAN Defense System

The 32nd USENIX Security Symposium

August 2023

Anaheim, California

Vulnerability Identification and Defense Construction in Cyber-Physical Systems

Presented to the School of Electrical and Computer Engineering

March 2023

University of Ottawa

Vulnerability Identification and Defense Construction in Cyber-Physical Systems

Presented to the School of Electrical Engineering and Computer Science

February 2023

American Uni. of Beirut

Protecting Against The Most Common CAN Bus Attacks

Presented to the Office of Naval Research (ONR)

October 2022

Purdue University

Demo: Attacks on CAN Error Handling Mechanism

Automotive and Autonomous Vehicle Security (AutoSec) Workshop

April 2022

Exposing New Vulnerabilities of Error Handling Mechanism in CAN

The 30th USENIX Security Symposium

August 2021

Evading Voltage-Based Intrusion Detection on Automotive CAN

The Network and Distributed System Security Symposium (NDSS)

February 2021

A Highly Portable CAN Bus Testbed

Presented to the Office of Naval Research (ONR)

January 2020

Purdue University

Exposing New Vulnerabilities of Error Handling Mechanism in CAN

Automotive Information Sharing and Analysis Center (Auto-ISAC)

June 2021

Advised Graduate Students

Current Zhaozhou Tang

Georgia Tech

Mentored Student Interns

2025 Hafsa Farhan

UDST

2025 Mohamed Elost

CMU-Q

2025 Mohammed Faheem Zaidi

Qatar University

Academic Teaching Experience

CS 590: IoT/CPS Security

Invited by Dr. Z. Berkay Celik

Guest Lecturer

Spring 2020

CS 426: Computer Security

Invited by Dr. Dave Tian

Guest Lecturer

Fall 2022

CS 426: Computer Security

Invited by Dr. Z. Berkay Celik

Guest Lecturer

Spring 2023

CS 528: Network Security

Invited by Dr. Dave Tian

Guest Lecturer

Spring 2023

Academic and Professional Services

Technical Program Committee Member

The 26th International Symposium on Research in Attacks, Intrusions and Defenses (RAID)

2023

Technical Program Committee Member

The Inaugural ISOC Symposium on Vehicle Security and Privacy (VehicleSec), co-located with NDSS

2023

Technical Program Committee Member

The Artifact Evaluation Committee of the 18th European Conference on Computer Systems (EuroSys)

2023

Reviewer

IEEE Transactions on Information Forensics and Security (T-IFS) 2022

IEEE Transactions on Intelligent Transportation Systems (T-ITS) 2023

Subreviewer

IEEE Symposium on Security and Privacy (IEEE S&P) 2023

32th USENIX Security Symposium 2023

IEEE Transactions on Dependable and Secure Computing (T-DSC) 2022

30th USENIX Security Symposium 2021

The Network and Distributed System Security Symposium (NDSS) 2021

Certifications

2016 Certified Ethical Hacker (CEH) EC-Council

2013 Cisco Certified Network Associate (CCNA) Cisco

Vulnerability Reports

CERT's Vulnerability Information and Coordination Environment (VINCE)

Case: VU#720158

Controller Area Network Standard (CAN Bus), ISO-11898

January 2021

- *Passive Error Regeneration*: Could be exploited to launch an immediate denial of service (DoS) attack
- *Deterministic Recovery Behavior*: Could be exploited to launch a persistent denial of service (DoS) attack
- *Error State Outspokenness*: Could be exploited to identify message sources, their error states, and to map the network
- Also reported the vulnerabilities to Bosch, ISO, ANSI, and SAE
- Gave a talk to the Automotive Information Sharing and Analysis Center (Auto-ISAC) explaining the vulnerabilities

Technology Transfers

Smart Information Flow Technologies (SIFT)

July 2021

RAandomized Identifier Defense (RAID)

Provides protection against error-handling attacks on CAN systems

Siege Technologies

September 2021

DUET Attack

A CAN injection attack that evades detection by voltage-based intrusion detection systems (VIDS)

Languages

Arabic Full Proficiency

English Full Proficiency

French Intermediate Proficiency

Spanish Intermediate Proficiency